Lab program-38:

38. Design a C program to simulate SCAN disk scheduling algorithm.

Code:

```c
#include <stdio.h>
#include <stdlib.h>

int main(void) {
    int n, head, disk_size, direction;
    int seek_time = 0;

    printf("Enter the number of disk requests: ");
    scanf("%d", &n);
    if (n <= 0) return 0;

    int request_queue[n];
    printf("Enter the disk request queue:\n");
    for (int i = 0; i < n; i++)
        scanf("%d", &request_queue[i]);

    printf("Enter the initial position of the disk head: ");
    scanf("%d", &head);
    printf("Enter disk size: ");
    scanf("%d", &disk_size);
    printf("Enter direction (1 for right, 0 for left): ");
    scanf("%d", &direction);

    for (int i = 0; i < n - 1; i++)
        for (int j = i + 1; j < n; j++)
            if (request_queue[i] > request_queue[j]) {
                int temp = request_queue[i];
```

```c
                int temp = request_queue[i];
                request_queue[i] = request_queue[j];
                request_queue[j] = temp;
            }

    printf("\nSCAN (Elevator) Disk Scheduling:\n");
    printf("Head Movement Sequence: %d", head);

    int split = 0;
    while (split < n && request_queue[split] < head) split++;

    if (direction == 1) {
        for (int i = split; i < n; i++) {
            seek_time += abs(head - request_queue[i]);
            head = request_queue[i];
            printf(" -> %d", head);
        }
        if (head != disk_size - 1) {
            seek_time += abs(head - (disk_size - 1));
            head = disk_size - 1;
            printf(" -> %d", head);
        }
        for (int i = split - 1; i >= 0; i--) {
            seek_time += abs(head - request_queue[i]);
            head = request_queue[i];
            printf(" -> %d", head);
        }
    } else {
```

```c
    } else {
        for (int i = split - 1; i >= 0; i--) {
            seek_time += abs(head - request_queue[i]);
            head = request_queue[i];
            printf(" -> %d", head);
        }
        if (head != 0) {
            seek_time += head;
            head = 0;
            printf(" -> %d", head);
        }
        for (int i = split; i < n; i++) {
            seek_time += abs(head - request_queue[i]);
            head = request_queue[i];
            printf(" -> %d", head);
        }
    }

    printf("\nTotal Seek Time: %d\n", seek_time);
    printf("Average Seek Time: %.2f\n", (float)seek_time / n);
    return 0;
}
```

OUTPUT:

```
Enter the number of disk requests: 3
Enter the disk request queue:
12
34
45
Enter the initial position of the disk head: 45

SCAN (Elevator) Disk Scheduling:
Total Seek Time: 0
Average Seek Time: 0.00
```